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GUIDANCE DOCUMENT 5A: SECTOR-SPECIFIC GUIDANCE DOCUMENT ON CEMENT

Disclaimer: This guidance document is not legally binding and it is of an explanatory nature. Legal provisions of Union law take precedence over the contents of this document and should always be consulted. The authentic texts of the EU legal instruments are those published in the Official Journal of the European Union. This guidance was prepared by Commission services and does not represent an official position of the Commission. Only the Court of Justice of the European Union is competent to authoritatively interpret Union law.

VERSION HISTORY

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1 PREFACE

1.1 Structure and content of the guidance documents

This series of documents has been written to support stakeholders by explaining the requirements of the CBAM Regulation in a non-legislative language. This series of guidance documents should benefit a variety of stakeholders. A specific focus has been put on clarifying the **requirements for operators of installations producing CBAM goods outside of the EU for the definitive period.**

- **Guidance 1** provides an overview of the main changes to the CBAM methodology between the definitive period and the concluded transitional period. It also introduces the CBAM and an overview of the compliance cycle, roles and responsibilities, as well as milestones and deadlines for operators of installations outside the EU during the definitive period and explains the general exemptions from the CBAM.

- **Guidance 2** provides a quick guidance for the intended reader of this document, the operator of an installation producing CBAM goods. It provides a roadmap of the most important concepts of CBAM emissions monitoring and where to find more information in this series of documents.

- **Guidance 3** sets out the monitoring and reporting obligations and recommendations which are potentially applicable to any affected producer of CBAM goods.

- **Guidance 4** provides an overview of how the adjustment for free allocation is to be determined and what information is needed.

- **Guidance documents 5a, 5b, 5c, 5d, 5e and 5f** present an overview per sector of the production processes and value chains for goods that are included in the scope of the CBAM, as well as sector-specific monitoring and reporting considerations for each CBAM good where this is relevant. These sections are supplemented by examples for each sector.

The series of guidance documents is accompanied by an electronic template for information that should be used by installation operators to communicate information to the authorised CBAM declarants and CBAM Registry.

Presentation of numbers in EU documents

To align with EU legal documents, this guidance document uses the following convention when presenting numbers.

The decimal separator used to separate the integral part of a number from its fractional part is a comma, e.g.: 0,890

Thousands, and powers of 10^{3n} thereafter, are separated by a space, e.g.:

- fifteen thousand is written as 15 000
- fifteen million is written as 15 000 000

1.2 Reference regulations and legal acts

Several regulations and legal acts are referred to throughout this series of guidance documents. For ease of reading and to avoid ambiguity, simplified references are used when citing these acts.

- **CBAM Regulation**: Regulation (EU) 2023/956 and Regulation (EU) 2025/2083 amending Regulation (EU) 2023/956. Notably, where article numbers are given in this document without further specification, they always refer to the CBAM Regulation.

  It can be accessed at the following link (consolidated version):
  http://data.europa.eu/eli/reg/2023/956/2025-10-20

- **Methodology Act:** Commission Implementing Regulation (EU) 2025/2547, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2547/oj

- **Default Values Act**: Commission Implementing Regulation (EU) 2025/2621, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation", also referred to as "default values determined in accordance with Annex IV to the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2621/oj**

- **Verification Principles Act**: Commission Implementing Regulation (EU) 2025/2546, in legal texts referred to as "Implementing act pursuant to Article 8(3) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2546/oj

- **Accreditation & Verification Requirements Act**: Commission Delegated Regulation (EU) 2025/2551, in legal texts referred to as "Delegated act pursuant to Article 18(3) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_del/2025/2551/oj**

- **Free Allocation Adjustment Act**: Commission Implementing Regulation (EU) 2025/2620, in legal texts referred to as "Implementing act pursuant to Article 31(2) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2620/oj

For acronyms and definitions used in this document, please see **Annex B** and **Annex A**.

A series of icons are used throughout to help guide the reader:

Icon	Description of use
	Points to information of particular importance for operators of installations producing CBAM goods.
Simplified!	Highlights simplified approaches of the general requirements of the CBAM.
	Used where recommended improvements are presented.
	Used where other documents, templates or electronic tools are available from other sources.
	Points to examples given for the topics discussed in the surrounding text.

2 CBAM GOODS AND PRODUCTION ROUTES

This section provides guidance on industry-sector-specific rules that apply for the definitive period (from 2026 onwards), for the cement sector. It deals with the specification of products covered by the CBAM and the relevant production routes.

2.1 Foreword to sector-specific sections

The following sections provide an overview of the different production routes for the goods under cement sector in CBAM and provide sector specific guidance. Where goods to which the same functional unit applies are produced in one installation via different production routes, the embedded emissions shall be determined in accordance with the averaging rules applicable in the definitive period.

Additional information on the production processes of the goods can also be found in the BREF[1] documents for best available techniques (BAT).

Diagrams used in the following sections.

For the system boundary graphics presented in the sections below, the **following conventions** are applied:

- **Production processes** (for which monitoring of the direct emissions would take place) are shown as **rectangles with white background**.
- **Optional** processes (e.g. CCS/CCU) are shown in **blue** boxes. In particular, CCS/CCU would not be taken into account for developing default values, but where you (as an operator) use them, the related emissions or emission savings should be taken into account for determining actual embedded emissions in line with the CBAM Regulation and the Methodology Act.
- **Materials** which are considered to have no embedded emissions are shown in **pink** boxes, materials with embedded emissions (precursor materials and final products, i.e. goods under the CBAM) in **green** boxes. Simple goods are shown in **normal** font, complex goods in **bold** font. Materials which are **optional** are shown in boxes with a **blue dashed** outline and must be included where actually used at the installation.
- Input materials are presented without trying to be complete. This means that the focus is on materials which are relevant for demonstrating the differences between different production routes. As a consequence, less important input materials and in particular fuels are usually omitted in order to keep graphs simple.
- **Electricity** as input is shown only in cases where it is the main "precursor" of the process (i.e. in particular for electric arc furnaces and electrolysis processes). Electricity is shown in **light grey** boxes.

[1] BAT Reference document (BREF), BAT being "Best Available Techniques" as defined by the IED (Industrial Emissions Directive). Relevant BREF documents are those for: the production of cement; for iron and steel production; large volume inorganic chemicals (which includes fertilisers); for Chlor-alkali; and for non-ferrous metals (which includes both aluminium and ferro-alloys). All BREFs can be found at the European IPPC Bureau, under https://eippcb.jrc.ec.europa.eu/reference.

2.2 Cement sector goods and production routes

- The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I,** Table 1 "Mapping of CN codes to aggregated goods categories" of the Methodology Act.
- **Annex I**, points 3.2 – Calcined clay, 3.3 – Cement clinker, 3.4 – Cement and 3.5 – Aluminous cement, which define the functional unit, system boundaries and possible precursors for cement-sector goods.
- **Annex II**, point B.9.2 "Additional rules for emissions from cement clinker production", providing sector-specific rules for monitoring and calculation of emissions for cement clinker.

2.2.1 Unit of production and embedded emissions for industry sector

The quantity of declared cement goods imported into the EU must be expressed in the production unit applicable to each cement-sector CBAM good in line with Article 4(5) of the Methodology Act. You must record the quantity of CBAM good(s) produced by the installation or production process(s), for the purposes of reporting.

Industrial sector	Cement
Production unit of goods	Tonnes (metric), reported separately for each type of CBAM good produced, by the installation or production process in the country of origin. Tonnes (metric) for calcined clay, cement clinker and aluminous cement, and tonnes of clinker content for cement goods (i.e., CN codes 2523 10 00, 2523 21 00, 2523 29 00 and 2523 90 00), reported separately for each type of CBAM good produced, by the installation or production process in the country of origin, in line with Annex I, points 3.2 to 3.5 of the Methodology Act.
Associated activities	Producing cement clinkers and calcined clays, grinding and blending cement clinker to produce cement.
Relevant greenhouse gas emissions	Carbon dioxide (CO_2)
Direct Emissions	Tonnes (metric) of CO_2e
Indirect Emissions	Quantity of electricity consumed (MWh), source and emissions factor used to calculate the indirect emissions in tonnes (metric) of CO_2 or CO_2e. Indirect emissions resulting from electricity consumption in the relevant production process shall be monitored and taken into account in the embedded emissions of cement CBAM goods, in accordance with Article 3(2) and Annex IV to the CBAM Regulation and with the Methodology Act

Industrial sector	Cement
Unit for embedded emissions	Tonnes CO_2e emissions per tonne of good, reported separately for each type of CBAM good, by the installation or production process in the country of origin. Tonnes CO_2e emissions per tonne of calcined clay, cement clinker or aluminous cement, and tonnes CO_2e emissions per tonne of clinker content for cement (excl. aluminous cement), reported separately for each type of CBAM good, by the installation or production process in the country of origin.

The cement sector has to account for both direct emissions and indirect emissions, in the definitive period, since cement goods are not listed in Annex II ("Goods for which only direct emissions are taken into account") to the CBAM Regulation and are therefore subject to the general rule in Article 3(2) of that Regulation on the inclusion of indirect emissions. Both direct and indirect emissions must therefore be taken into account in the embedded emissions of cement-sector CBAM goods, with direct and indirect components determined separately in line with the Methodology Act.

For cement (excl. aluminous cement), embedded emissions are determined per tonne of clinker content, and where cement is produced and placed on the market with different clinker contents, emissions shall be calculated and averaged in line with the composition-based rules set out in Annex III to the Methodology Act.

Emissions must be reported in metric tonnes of CO_2 equivalent (t CO_2e) emissions, per tonne of good output. This figure must be calculated for the specific installation or production process in your country of origin.

The following sections set out how the system boundaries of cement sector goods should be defined and identify elements of the production processes that must be included for the purposes of monitoring and reporting.

2.2.2 Definition and explanation of goods covered

The following Table 2-1 lists the relevant goods in scope for the CBAM definitive period in the cement industry sector. The CN/TARIC codes in the central column are the starting point for which a 'production process' is to be identified for the purpose of monitoring emissions, encompassing all production routes used within the installation in line with Annex I to the Methodology Act.

Table 2-1: CBAM goods in the cement sector[2]

Aggregated goods category	CN/TARIC Code	Description
Calcined clay	2507 00 80 80[3]	Other kaolinic clays, calcined
Cement clinker	2523 10 00[4]	Cement clinkers
Cement	2523 21 00	White Portland cement, whether or not artificially coloured
	2523 29 00	Other Portland cement
	2523 90 00[5]	Other hydraulic cements
Aluminous cement	2523 30 00	Aluminous cement[6]

The aggregated goods categories listed in Table 2-1 include both finished cement goods and precursor goods (intermediate products) that are consumed in the production of cement.

The system boundaries specified in the Methodology Act do not mention explicitly which precursors are relevant for the production process. Therefore, Table 2-2 should help to identify the possible precursors by aggregated goods category and production route.

Table 2-2: Aggregated goods categories, their production routes and possible precursors

Aggregated Goods Category ***Production route***	**Precursors**
Calcined clay	None
Cement clinker	None
Cement	Cement clinker; calcined clay (if used in the process).
Aluminous cement	None

Precursor goods relevant to the system boundary are 'cement clinker[7]' (CN code 2523 10 00), which includes both white clinker (used to make white cement) and grey clinker, and 'calcined clay' (TARIC code 2507 00 80 80), which is a clinker substitute and may be used to modify the properties of the cement produced.

2 Source: Annex I to the CBAM Regulation and Annex I ("Mapping of CN codes to aggregated goods categories") to the Methodology Act.

3 Non-calcined clays are found under CN code 2507 00 80, but they are excluded from the CBAM scope.

4 TARIC codes 2523 10 00 10 (White cement clinker) and 2523 10 00 90 (Other cement clinkers including grey clinker) apply.

5 TARIC codes 2523 90 00 10 (White hydraulic cement) and 2523 90 00 90 (Other hydraulic cements including grey hydraulic cement) apply.

6 Also referred to as 'Calcium Aluminate Cement'.

7 Distinction is made between grey and white clinker through the TARIC codes to be applied.

These precursors are defined as simple goods, as they cannot be produced out of other CBAM goods.

The finished cement goods listed in Table 2-1 comprise both white Portland cement, grey Portland cement, other hydraulic cements and aluminous cement. These goods are defined as complex goods (with the exception of aluminous cement) as they include the embedded emissions from precursor goods in accordance with the definitions of simple and complex goods in the CBAM Regulation and the definition of precursors in Article 1 of the Methodology Act.

Other constituents used in cement manufacture, in particular granulated blast furnace slag, fly ash and natural pozzolana that are used in the manufacture of other hydraulic cement goods (including blended or 'composite' cements) are not CBAM goods and therefore are not in scope for the CBAM as precursors. Their use and the associated emissions are taken into account through the direct and indirect emissions of the cement production process, but no additional embedded emissions of these constituents are added as separate precursors under the CBAM rules.

Cement sector goods are produced by a number of different production process and routes, outlined below.

2.2.3 Definition and explanation of relevant production processes and routes

The system boundaries of precursors and cement goods are distinct and may, under certain conditions, be added together to include all processes directly or indirectly linked to the production processes for these goods, including input activities to the process and output activities from the process.

2.2.3.1 Calcined clay production process

Calcined clay may be used as a clinker substitute. Kaolinic clay that is calcined (metakaolin) can be added to cement in place of clinker in varying proportions in order to modify the properties of the cement mixture.

The Methodology Act, in particular Annex I, point 3.2 ("Calcined clay"), sets out the functional unit and system boundaries for the calcined clay production route. For the purposes of this guidance, the system boundaries for direct emissions monitoring can be described as encompassing the following:

> *" – All processes directly or indirectly linked to the production processes, such as raw material preparation, mixing, drying, and calcining, and flue gas cleaning.*
>
> *– CO_2 emissions from the combustion of fuels as well as from raw materials, where relevant."*

There are no precursors for this production process. Indirect emissions that result from electricity consumed by the production process shall also be determined and taken into account in the embedded emissions of calcined clay, in accordance with Article 3(2) of the CBAM Regulation and the rules on indirect emissions and system boundaries in the Methodology Act.

Note that CN code 2507 00 80 includes both calcined and non-calcined kaolinic clays. As a result of simplification, only calcined kaolinic clays (TARIC code 2507 00 80 80) are included in the scope of CBAM in the definitive phase. Simplified

2.2.3.2 Cement clinker production process

Cement clinker is produced in clinker plants (kilns) by the thermal decomposition of calcium carbonate to form calcium oxide, followed by the clinkering process in which the calcium oxide reacts at high temperatures with silica, alumina and ferrous oxide to form a clinker. Grey and white clinkers may be produced depending on the temperature of the process and purity of raw materials.

The Methodology Act, in particular Annex I, point 3.3 and Annex II, point B.9.2, sets out the functional unit, system boundaries and specific rules for monitoring emissions from cement clinker production. For the purposes of this guidance, the system boundaries for direct emissions monitoring can be described as encompassing:

> *" – Calcination of limestone and other carbonates in the raw materials, conventional fossil kiln fuels, alternative fossil-based kiln fuels and raw materials, biomass kiln fuels (such as waste-derived fuels), non-kiln fuels, non-carbonate carbon content in raw materials, or alternative raw materials such as fly ash used in the raw meal in the kiln and raw materials used for flue gas scrubbing."*

Typical raw materials include limestones and shales. Direct emissions of the cement clinker production process result from the combustion of both kiln and non-kiln fuels and from raw materials used in the process such as limestone. In line with the Methodology Act (in particular Annex I, point 3.3 and Annex II, point B.9.2), direct emissions include both process emissions from the calcination of carbonates in the raw materials and combustion emissions from all relevant fuels used in the kiln system and for drying of raw materials and fuels.

Indirect emissions that result from electricity consumed by the production process shall also be determined and taken into account in the embedded emissions of cement clinker, in accordance with Article 3(2) of the CBAM Regulation and the rules on indirect emissions and system boundaries in the Methodology Act.

In line with the above definition of system boundaries, the following production steps may be regarded as being within the system boundaries of cement clinker installations:

- Raw material preparation – grinding, milling, homogenisation.
- Fuel storage and preparation – for conventional and waste derived fuels.
- Clinker production ('clinker burning') – all steps for the integrated kiln system including preheating, kiln processing and clinker cooling.
- Intermediate storage – storage of cement clinker under cover before export off site or cement grinding.
- Emissions control – for treating releases to air, water or ground.

A variation on the clinker production process may be with permanent geological storage i.e. carbon capture and sequestration (CCS). Where CO_2 is captured and permanently stored or used in accordance with the CBAM rules, this should be reflected in the determination of embedded emissions in line with the Methodology Act; however, CCS/CCUS is not taken into account when establishing default values.

Note that distinction is made between white clinker and other clinker (incl. grey) used in the production of cement.

2.2.3.3 Cement production process

Cement (apart from aluminous cement) is defined as a complex good as it is produced from precursors cement clinker and possibly calcined clay.

Cement is produced in a grinding plant (cement mill), which may be located at the same installation that produced the cement clinker, or at a separate standalone plant. Cement clinker is ground and blended with certain other constituents to produce the finished cement product. Depending on the mix of different constituents this may be Portland cement, blended cement (containing a mix of Portland cement and other hydraulic constituents) or other hydraulic cements.

The Methodology Act, in particular Annex I, point 3.4, sets out the functional unit and system boundaries for cement production. For the purposes of this guidance, the system boundaries for direct emissions monitoring can be described as encompassing:

> *" – All processes directly or indirectly linked to the production processes, where relevant for drying of materials."*

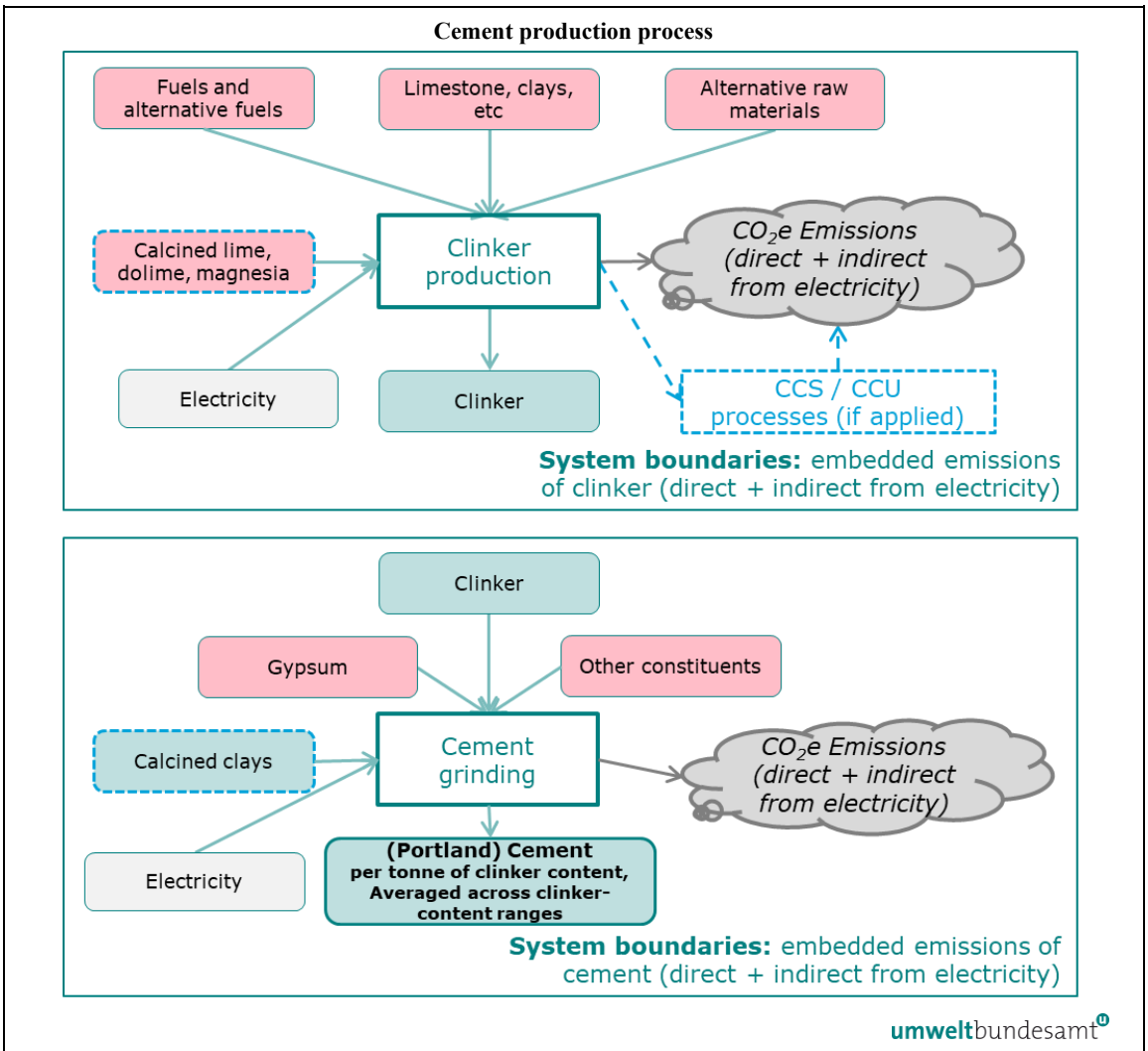
Precursors are cement clinker and calcined clay (if used in the process). Indirect emissions that result from electricity consumed by the cement production process shall also be determined and taken into account in the embedded emissions of cement, in accordance with Article 3(2) of the CBAM Regulation and the rules on indirect emissions and system boundaries in the Methodology Act.

In line with the above definition of system boundaries, the following production steps may be regarded as being within the system boundaries of cement installations:

- Material preparation – materials (cement clinker, calcined clay and mineral additives) handling and pre-treatment e.g. preheating and drying mineral additives.
- Cement production – all steps, including crushing, grinding, further milling and separation by particle size.
- Cement storage, packaging and dispatch.
- Emissions control – for treating releases to air, water or ground.

Figure 2-1 following shows how the cement clinker and cement production processes relate to each other.

Figure 2-1: *System boundaries of cement clinker and cement production processes.*



Notes to figure: In this figure, the green or blue boxes with a blue dashed outline (e.g. calcined clays) represent optional precursors or process steps that are only included where they are actually used at the installation. Note that embedded emissions (and therefore SEEs) need to be reported separately for white hydraulic cements and for other hydraulic cements (incl. grey).

For hydraulic cements, note that embedded emissions (and therefore SEEs) need to be reported separately for white hydraulic cements and for other hydraulic cements (incl. grey).

2.2.3.4 Aluminous cement production process

Aluminous cement is regarded as a simple good as it is produced directly from aluminous clinker by a continuous production process and is ground without the addition of further additives in line with the definition of simple goods and the aggregated goods category "Aluminous cement" in Annex I to the Methodology Act.

The Methodology Act, in particular Annex I, point 3.5, sets out the functional unit and system boundaries for the aluminous cement production route. For the purposes of this guidance, the system boundaries for direct emissions monitoring can be described as encompassing:

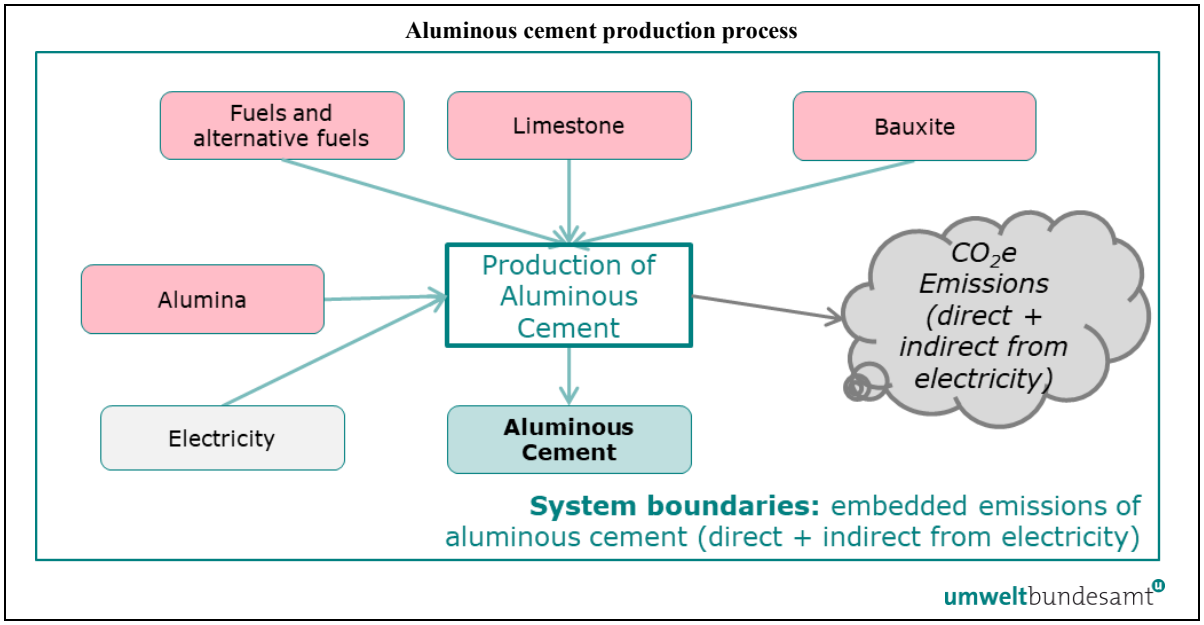
> *" – All processes directly or indirectly linked to the production process where fuel is combusted.*

– Process emissions from carbonates in raw materials, if applicable, and flue gas cleaning."

There are no precursors for this production process. Indirect emissions that result from electricity consumed by the production process shall be determined and taken into account in the embedded emissions of cement-sector CBAM goods, in accordance with Article 3(2) of the CBAM Regulation and the rules on indirect emissions in the Methodology Act.

In line with the above definition of systems boundaries, the integrated production of aluminous cement includes both clinkering and cement grinding production steps, from raw material preparation through to emissions control.

Figure 2-2: System boundaries of the aluminous cement production process



Note that alumina (produced from bauxite) is treated as a raw material for the purposes of the aluminous cement aggregated goods category and is not a precursor under Annex I to the Methodology Act. Its contribution to embedded emissions is taken into account through the direct and indirect emissions of the aluminous cement production process, and no separate embedded emissions of alumina are added as a precursor.

3 CEMENT SECTOR-SPECIFIC REQUIREMENTS

The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I, point 3** – Functional unit and system boundaries. Sub-sections 3.2 to 3.5 (cement sector aggregated goods categories).

- **Annex II**, **Section B** – Monitoring of direct emissions at installation level, sub-section **B.9.2**. Sector specific requirements, additional rules for process emissions from the cement sector, covering: **B.9.2.1** additional rule for Method A (input based); **B.9.2.2** additional rule for Method B (output based); **B.9.2.3** additional rule for emissions related to discarded cement kiln dust/bypass dust.
- **Annex IV, point 2** – Sector-specific parameters to be included in the operator's emissions report.

3.1 Sector-specific requirements for monitoring and reporting

Direct and indirect embedded emissions should be monitored in line with the methodology set out in the Methodology Act and outlined above.

3.1.1 Emissions covered

Relevant direct emissions that must be monitored and reported for the cement sector are:

- Carbon dioxide emissions (direct) from the fuel combustion process[8] from stationary plant only (excludes emissions from any mobile plant such as vehicles).
- Carbon dioxide emissions (direct) from the process, resulting from:
    - Thermal decomposition of carbonate containing raw materials (such as limestone, dolomite etc);
    - Non-carbonate carbon content in the raw materials (such as carbonaceous clay, limestone, shale);
    - Alternative raw material (such as fly ash used in the raw meal), or from any fossil/biomass additives used.
    - Discarded cement kiln dust (CKD) or bypass dust.
- Carbon dioxide emissions (direct) resulting from the production of measurable heating (e.g. steam) and cooling that is consumed within the system boundaries of the production process, regardless of the location of the production of the heat (i.e. from on-site generation or from imports from off-site).
- Carbon dioxide emissions (direct) resulting from emissions control (e.g. from carbonate raw materials such as soda ash used for acidic flue gas cleaning). This is included for any good where this is applicable.

Direct emissions from the different source streams above are not reported separately but are added together to result in the total direct emissions for the installation or production process.

Indirect emissions from electricity consumed must be reported separately from direct emissions.

[8] For both kiln and non-kiln fuels. Cement kiln fuels include conventional fossil fuels such as natural gas and coal, alternative fossil-based fuels such as petroleum coke or waste chipped vehicle tyres, and biomass fuels (biomass wastes). Non-kiln fuels refers to fuels used outside of kilns e.g. to calcine clays in flash calciners, and where cement materials are dried.

3.1.2 Additional rules

Determining process emissions

For determining direct emissions from cement clinker production, additional rules also apply for the monitoring of process emissions from raw meal components depending on whether the **activity data** refers to:

- Process **input** material (e.g. limestone) based on:
  - the carbonate content of the process input (calculation **Method A**); and
  - an adjustment is made for cement kiln dust (CKD) or bypass dust that leaves the kiln system.
- Process **output** material e.g. the amount of clinker produced (calculation **Method B).**

Note that both methods are considered equivalent, i.e. you, as the operator, should choose the method which leads to the more reliable data, which is more applicable for your equipment, and that avoids unreasonable costs.

Calculation of emissions related to discarded CKD or bypass dust

As the operator, you shall add CO_2 process emissions, from bypass dust or cement kiln dust (CKD) leaving the kiln system, corrected for a partial calcination ratio of CKD.

- Minimum requirements: An emission factor of 0,525 t CO_2/t dust is applied.



Recommended improvement**: The emission factor (EF) is determined at least once annually in line with the provisions of Methodology Act Annex II, point B.5.4. requirements for laboratory analyses and using **the following formula:

$$EF_{CKD} = \left(\frac{EF_{Cli}}{1+EF_{Cli}} \cdot d\right) / \left(1 - \frac{EF_{Cli}}{1+EF_{Cli}} \cdot d\right) \quad \text{(Equation 28)}$$

Where:

EF_{CKD} … Emission factor of partially calcined cement kiln dust [t CO_2/t CKD];

EF_{Cli} … Installation-specific emission factor of clinker [t CO_2/t clinker];

d … Degree of CKD calcination (released CO_2 as percentage of total carbonate CO_2 in the raw mix).

Method B - clinker output based

For this method, a sector-specific rule is given in the Methodology Act:

Activity data AD_j for clinker production [t] over the reporting period may be determined by either:



- Direct weighing of clinker (if technically feasible); or
- Based on cement deliveries, by material balance using the following stock adjustment calculation:

$$Cli_{prod} = (Cem_{deliv} - Cem_{SV}) \cdot CCR - Cli_s + Cli_d - Cli_{SV}$$ (Equation 27)

Where

Cli_{prod} is the amount of clinker produced expressed in tonnes,

Cem_{deliv} the amount of cement deliveries expressed in tonnes,

Cem_{SV} the cement stock variations expressed in tonnes,

CCR the clinker to cement ratio (tonnes clinker per tonne cement),

Cli_s the amount of clinker supplied expressed in tonnes,

Cli_d the amount of clinker dispatched, and

Cli_{SV} the amount of clinker stock variations expressed in tonnes.

As a minimum requirement to determine the emission factor, a standard value of 0,525 t CO_2/t clinker shall be applied. Recommended improvement would be to perform analyses of the clinker to determine the EF.

For the **conversion factor** CF_j (see Methodology Act Annex II, point B.3.1.2). It is allowed to always use the conservative assumption that CF_j = 1 in order to reduce monitoring efforts.

Clinker to cement ratio (CCR)

In calculating the embedded emissions in cement goods, the majority of emissions result from cement clinker. Therefore, the CCR, which is the mass ratio of tonnes cement clinker consumed per tonne of cement produced (also known as the 'clinker factor'), must be taken into account. For CN codes 2523 10 00, 2523 21 00, 2523 29 00 and 2523 90 00, the functional unit is the tonnes of clinker contained in the goods. For the other CN codes, the default functional unit is the tonnes of product.

The CCR should be derived either:

- Separately for each of the different cement products, based on laboratory analyses in line with the provisions of Annex II point B.5.4; or
- By calculation, as ratio from the difference of cement deliveries and stock changes and all materials used as additives to the cement including by-pass dust and cement kiln dust.

The CCR is expressed as a percentage (%) value, which typically ranges from 80 up to 95 % for Portland cement. The CCR is particularly relevant for the calculation of relevant embedded emissions for blended or composite cements that are produced, where the clinker content can range widely for different types of composite cement[9] with the balance

[9] European standard EN 197-1 defines five main common cement types CEM I (Portland cement) to V (Composite cement) and 27 different product types, where the clinker content in blended and composite cements (CEM II to V) may range from 95% down to as low as 5-20%.

being made up of other constituents such as mineral additives,[10] with zero embedded emissions.

For cement goods sold in different ranges of composition, you as the operator will need to calculate the specific embedded emissions of the goods according to the tonnes of clinker contained in the goods, using the following equation.

$$SEE_{g(cki)} = SEE_g \cdot CK_i \qquad \text{(Equation 64)}$$

Where:

$SEE_{g(cki)}$ is the specific embedded emissions of the good with clinker content CK_i expressed as t CO_2e per tonne of cement;

SEE_g is the specific embedded emissions calculated in Equation 59 or 62 (i.e. expressed as t CO_2e per tonne of clinker), and

CK_i is the average clinker content of the goods of a given composition in tonnes of clinker per tonnes of goods.

Note that for cement goods that are sold in custom-made compositions at the request of the client, you as the operator are required to issue a declaration of clinker content for each consignment of the good; and in substituting into equation 64, *CKi* will be the specific clinker content of the consignment.

White clinker vs. other clinkers (incl. grey)

For cement clinker, TARIC codes 2523 10 00 10 (White cement clinker) and 2523 10 00 90 (Other cement clinkers including grey clinker) apply. In practice, embedded emissions (and therefore SEEs) for clinker goods need to be calculated separately for white cement clinker and for other cement clinkers. This also applies to hydraulic cements, for which TARIC codes 2523 90 00 10 (White hydraulic cement) and 2523 90 00 90 (Other hydraulic cements including grey hydraulic cement) apply.

3.1.3 Additional reporting parameters

The following Table 3-1 lists out the additional information that needs to be provided by you as an operator to importers, in your verified operator's emissions report and a summary thereof to them, as set out in Methodology Act Annex IV, Part 2 (Sector-specific parameters to be included in the emissions report).

Table 3-1: Additional cement sector parameters requested in the CBAM declaration

Aggregated goods category	Reporting parameter
Calcined clay[11]	– N.a. (not applicable)
Cement clinker	– N.a. (not applicable)

[10] Mineral additives (mainly gypsum) along with secondary mineral additives (blast furnace slag and fly ash) are not treated as precursors for the purposes of calculating embedded emissions for cement goods and are therefore treated as having zero embedded emissions in that calculation.

[11] Note that CN code 2507 00 80 includes non-calcined kaolinic clays. Only calcined clays under TARIC code 2507 00 80 80 are to be reported.

Cement	– Clinker content of cement. This is the: – Mass ratio of tonnes cement clinker consumed per produced tonne of cement (clinker to cement ratio expressed in per cent);
Aluminous cement	– N.a. (not applicable)

You need to ensure that you collect all the parameters necessary for your CBAM goods and communicate them to the importers of your goods. The importer will need to report the additional parameters in their annual CBAM declaration for the relevant reporting period.

3.2 Example for splitting a cement installation into separate production processes

In defining the system boundary for a production process, you as an operator will need to decide what physical production units belong to the production process(es) and what inputs, outputs, and emissions are relevant. An example of this is given as shown in the Table 3-2 below for the cement sector.

For a notional cement plant that both produces and exports cement clinker (CN 2523 10 00) and cement (CN 2523 29 00), the operator would need to go through the following steps to split the cement plant into separate production processes under the CBAM:

Step 1: List all goods, physical units, inputs, outputs and emissions to/from the installation.

In this first step, the operator uses the existing information available on their installation, such as lists of industrial equipment and plans, to identify the:

- Physical units carrying out the production processes at their installation, such as kilns, boilers, dryers, flue gas cleaning, ball mills, bagging plant.
- Process inputs required for making the goods e.g. raw materials, fuel, electricity.
- Outputs from the process e.g. goods produced, by-products, heat, waste gases.
- Emissions from the process.

These are then listed out in Table 3-2 below. Note that emissions caused by the "*purchase and maintenance of infrastructure and equipment*" are to be excluded from this listing exercise.

Table 3-2: Checklist for inputs, physical units, outputs and emissions for an example cement installation.

Inputs	Physical units	Outputs	Relevant CBAM Emissions
Kiln – fossil fuels[12] e.g. coal, HFO[13] Kiln – alternative and waste fuels (to cement clinker kiln) e.g. high CV fraction of MSW[14] Kiln – electricity consumed by clinker kiln and associated equipment Mill – fossil fuels to cement dryer Mill – electricity consumed by cement grinding plant and associated equipment Kiln – raw materials[15]: Limestone, clays Kiln – alternative raw materials: e.g. fly ash Mill – cement clinker from kiln Mill – additives used in cement manufacture	Kiln system and associated equipment e.g. for raw meal preparation Mill – grinding equipment (including dryer) and associated plant e.g. for bagging cement Other industrial equipment unrelated to cement production (to be excluded from system boundaries). Heat exchanger for district heating Flue gas cleaning equipment (for treating gaseous and dust emissions)	Kiln – cement clinker[16] Mill – cement goods, by type[17] Kiln – other outputs[18]: e.g. cement kiln dust District heating, (or cooling or electricity)[19]	Kiln – direct emissions from combustion of fuels Kiln – direct emissions from alternative and waste fuels Kiln – indirect emissions from electricity consumed. Kiln – direct process emissions from carbonates Mill – indirect emissions from electricity consumed.

12 Fuels combusted to generate heat for the use in the process under consideration or elsewhere. Both the fuel quantity (and in particular its carbon content / emission factor) as well its energy content are relevant for attributing it to different production processes.

13 Heavy Fuel Oil.

14 High calorific value fraction of municipal solid waste.

15 Raw materials are materials which participate in other chemical reactions or which are physically modified in the process for generating the product.

16 Precursor or intermediate good or product: where the production process also includes a 'finished' good. The precursor product may also be an output from the installation; for example, if the operator exported both cement clinker and cement from the installation.

17 Cement finished good – physical product output of the installation/production process monitored.

18 Other products (by-products) and waste: only need to be monitored if relevant in terms of carbon content for the determination of the production process's emissions, and energy content for corroboration purposes.

19 Measurable heat (or cooling or electricity, if fuels are used in their generation) exported from a CBAM installation or production process should be treated like a second product, i.e. a certain amount of emissions must be subtracted from the emissions of that production process.

Step 2: Identify relevant production processes and production routes.

In this step, the operator identifies that the installation produces cement clinker and cement, each of which is an aggregated goods category listed in Annex I, Table 1 ("Mapping of CN codes to aggregated goods categories") of the Methodology Act.

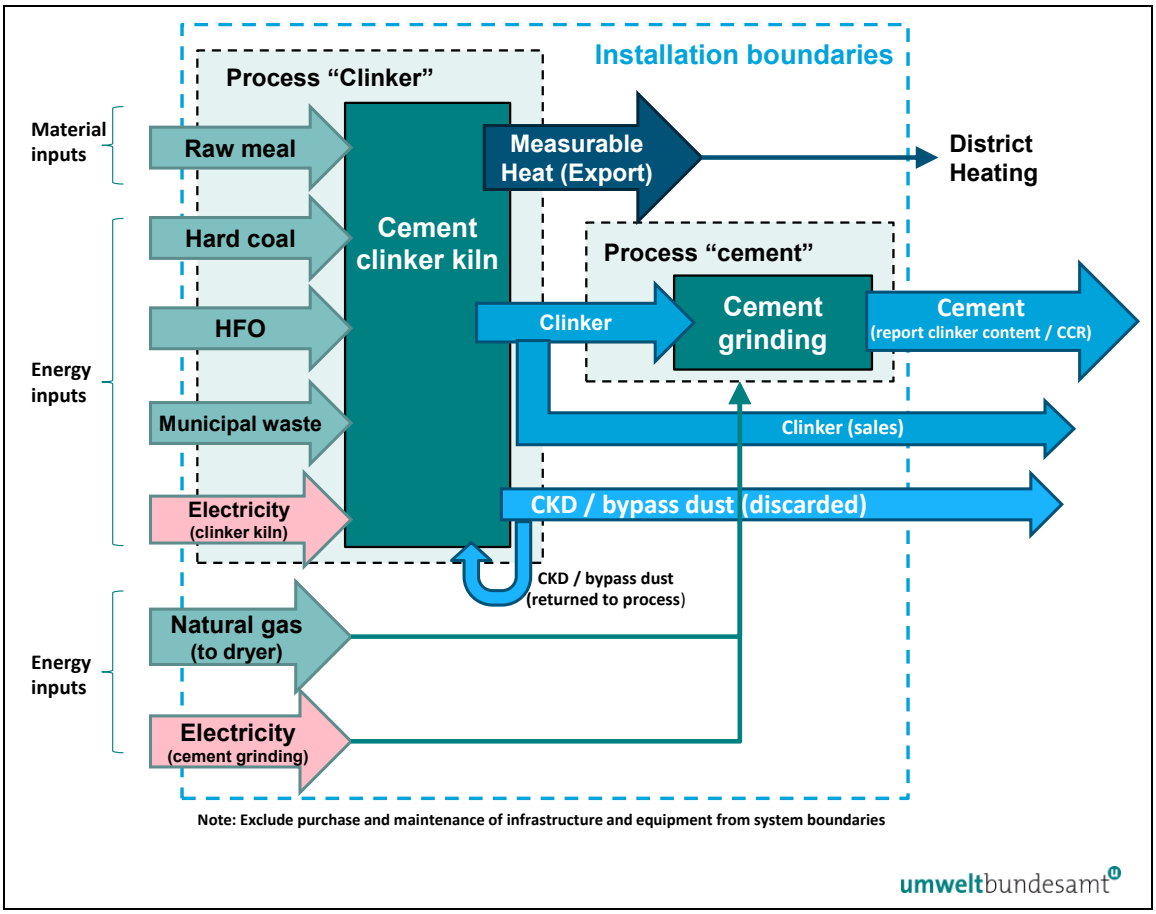
Each aggregated goods category defines system boundaries to be considered when defining which CN codes or groups of CN codes define a 'production process' for the purpose of monitoring, covering goods to which the same functional unit applies and, where relevant, encompassing all production routes used within the installation. The operator uses the Table 3-2 as a checklist for assigning the relevant inputs, outputs and emissions to each production processes. This is relatively simple in most cases, for example:

- Cement clinker (CN 2523 10 00) production process:
    - Physical units: Cement kiln, including preheaters, pre-calciner, clinker cooler, and associated auxiliary equipment such as flue gas cleaning.
    - Inputs / sources streams: Fuels, electricity, raw materials and alternative raw materials for the process.
    - Outputs (goods): Cement clinker, kiln dust (re-introduced it into the clinker production process).
    - Other outputs: measurable heat exported to district heating network.
    - Emission sources: direct (combustion and process) and indirect emissions (electricity consumed) related to the kiln system.
- Cement (CN 2523 29 00 and 2523 90 00)[20] production process:
    - Physical units: Grinding plant, direct fired dryer and associated auxiliary equipment e.g. cement bagging plant.
    - Inputs / source streams: Cement clinker, electricity, fuels for the dryer, additive materials used in making cement such as gypsum.
    - Outputs (goods): Cement.
    - Emission sources: direct (from cement dryer, if applicable) and indirect emissions (from electricity consumed) related to the grinding process.

Using a schematic helps to visualise the respective system boundaries of each production process and production route, and to assign inputs, outputs and emissions accordingly.

[20] For this example it is assumed that grey Portland cement and other hydraulic cements are produced.

Figure 3-1: *Schematic used to define the system boundaries of an example cement clinker and cement production processes*



In the case of the above cement installation, both the kiln system and cement grinding plant are relatively self-contained parts of the installation, without shared equipment, and there is no doubt about the system boundaries of each production process. The only element that is not widely found in this industry is the heat recovery from the clinker kiln for the purpose of district heating. In practice it would not be a separate production process, but in calculating the attributed emissions of the clinker production process, the heat would be taken into account.

The following worked example for the cement sector shows how, for the production processes defined, relevant emissions are calculated and assigned to production processes, and specific embedded emissions are calculated. For simplification, the district heating is omitted in that example, as are the additional direct emissions from the dryer before the cement grinding.

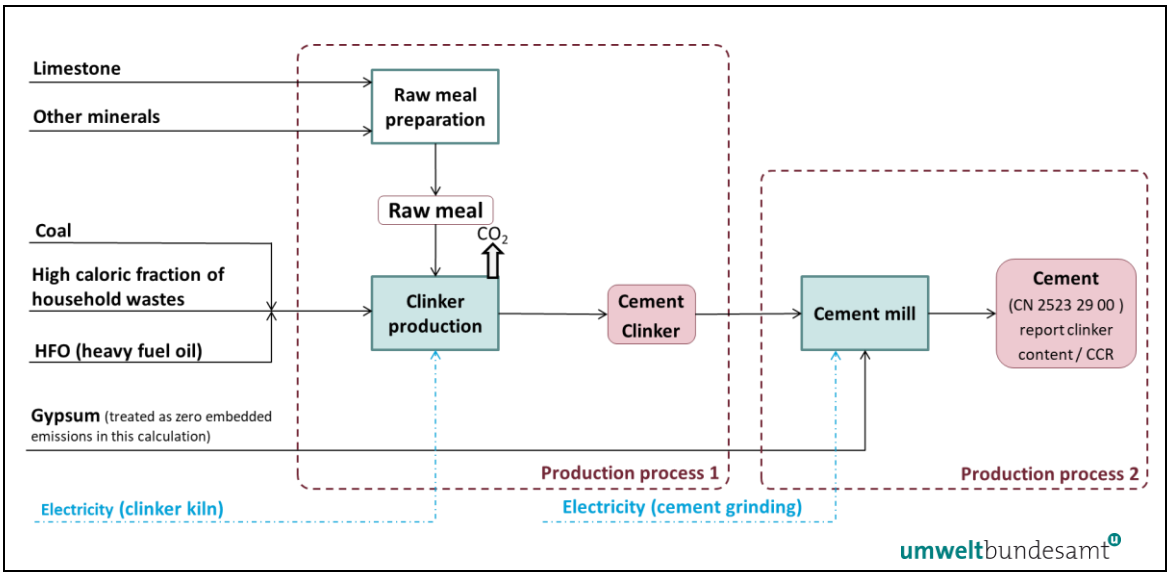
4 WORKED EXAMPLE FOR THE CEMENT SECTOR

The following worked example shows how specific embedded emissions are derived for cement sector goods. The resulting embedded emissions of imports into the EU are then calculated at the end of the example for reporting by the importer in their annual CBAM declaration for the relevant reporting period.

In this example the installation produces two products, cement clinker and cement, each of which is treated as a separate production process, as these goods refer to separate CN codes and each is a separate category of CBAM aggregated good. The example refers to clinker and cement in general without specifying whether the goods are grey or white clinker/cement. The example is thus illustrative of both cases.

Figure 4-1 gives an outline view of the installation and shows the system boundaries as a hatched line for each production process. The physical units carrying out each production process have been grouped under 'Clinker production' and 'Cement mill' and the different inputs and outputs and sources of emissions have been identified for each production process. Note that for the cement sector, indirect emissions are to be included, and therefore electricity is included in the calculation for this example.

Figure 4-1: Cement example – Overview



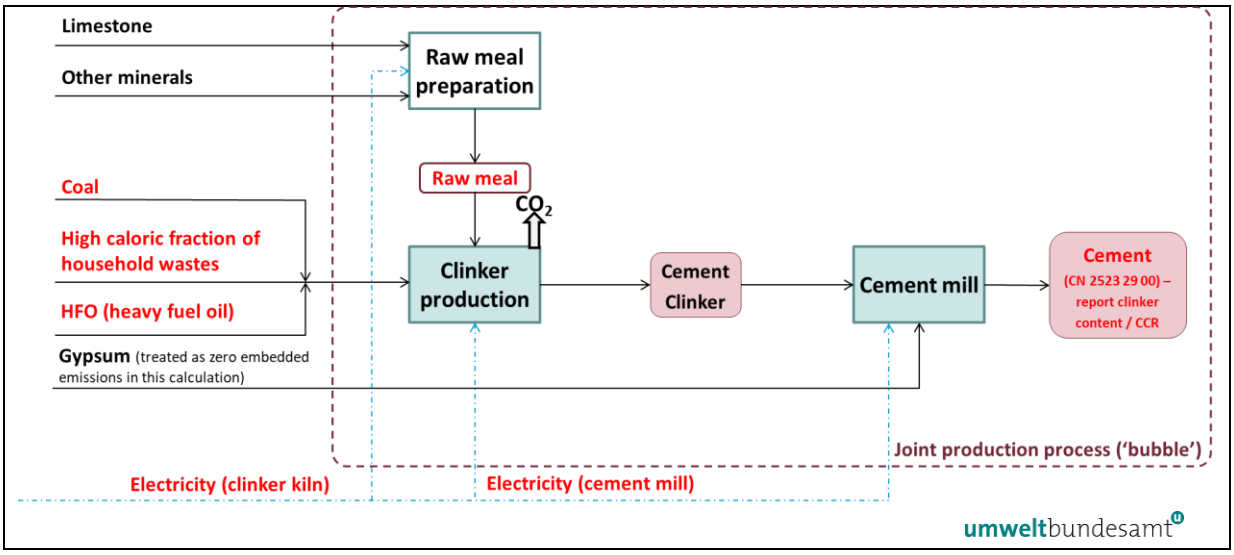
The two relevant production processes defined above are:

- Production process 1 – cement clinker produced in a cement kiln. The system boundaries of this production process have been defined as including inputs of raw materials (limestone and other minerals), fuels (coal, heavy fuel oil (HFO) and fractions of household waste) and electrical energy. Output from the process is cement clinker that is a precursor for production process 2.

- Production process 2 – cement produced in a cement mill. The system boundaries of this production process have been defined as including the inputs gypsum (which as a raw material is not a precursor (not included in Annex I to the CBAM Regulation), and therefore treated as having zero embedded emissions for the purposes of this calculation), precursor cement clinker (which has embedded emissions) and electrical energy. Output from the process is cement (with a need to report clinker content / CCR).

In this case, where all the output from production process 1 of precursor cement clinker goes directly into production process 2, it is possible to define a joint production process, or '**bubble**', whereby the system boundaries of the production processes are combined, as shown in *Figure 4-2*.

Figure 4-2: Cement example – joint production process ('bubble approach') and complete monitoring approach – red elements indicate inputs/outputs/activity data to be monitored within the combined system boundary.



The system boundary has been redrawn to encompass both production processes previously defined for each CBAM aggregated good.

The inputs and outputs highlighted in red text above are the parameters that would need to be monitored by the operator in order to attribute emissions and determine direct and indirect specific embedded emissions within the joint production process ('bubble').

The direct and indirect emissions that are monitored in this example result from:

- Direct emissions from fuel combustion – fossil fuels (coal and HFO) and from a mixed fossil/biomass fuel derived from household waste (an alternative fuel).
- Direct emissions from the process – resulting from the thermal decomposition of carbonates in the raw meal input (produced from limestone and other minerals) to the cement kiln system.
- Indirect emissions from the electrical energy consumed by the joint production process.

The activity level of cement also needs to be monitored in terms of the tonnes of clinker contained in the cement goods. It appears that the monitoring becomes simpler with this 'bubble approach". However, as the clinker amount is the functional unit for the cement produced, its quantity (as well as the quantity of other inputs used such as gypsum) need to be monitored for each type of cement produced. The associated embedded emissions of clinker do not have to be monitored separately but are the result of the overall bubble approach, divided by the total amount of clinker produced. Similarly, there is no need to split quantities of electrical energy consumed according to the two processes.

The following tables summarise the fuel, electrical energy and raw material inputs monitored for determining specific embedded emissions (SEE). The calculation of the SEE values is carried out in two steps:

- Step 1 – derives SEE values for the functional unit cement clinker; and

- Step 2 – derives SEE values for cement, taking into account i) embedded emissions of the bubble production process, and ii) the clinker to cement ratio (CCR) of each type of cement produced, to determine the tonnes of clinker contained in the cement goods.

Note that if any cement clinker produced by the installation were to be diverted and separately sold, the embedded emissions calculated under step 1 would have to be determined separately (without indirect emissions of the cement grinding plant) and would also need to be communicated by the operator to the purchaser of the cement clinker goods. In that case the 'bubble approach' would not be allowed.

In the following, we demonstrate both approaches, calculation with bubble approach and without, i.e. using two separate production processes.

4.1 Calculation using separate processes

Table 4-1: Calculation of direct and indirect emissions, and SEE values for cement clinker

Direct emissions	**AD (t)**	**NCV (GJ/t)**	**EF (t CO_2/t or t CO_2/TJ)**	**Biomass %**	**Emissions fossil (t CO_2)**	**Emissions biomass[21] (t CO_2)**
Process emissions						
Cement clinker (standard factor)[22]	1 255 000		0,525		658 875	
Combustion emissions						
Coal	88 000	25	95		209 000	0
High NCV household waste[23]	25 000	20	83	15%	35 275	6 225
HFO	43 000	40	78		134 160	0
Total direct emissions					**1 037 310**	
Indirect emissions	**AD (MWh)**		**EF (t CO_2 / MWh)**		**Emissions (t CO_2)**	
Electricity consumed	81 575		0,626		51 066	
Clinker production (tonnes)	1 255 000					
Step 1: Specific embedded emission (SEE) values are derived using direct and indirect emissions and activity data for cement clinker.						
Cement clinker	**Direct**	**Indirect**				
SEE	**0,82654**	**0,04069**	**t CO_2 / t**			

[21] Zero-rated biomass, i.e. compliant with the RED II criteria, where applicable.

[22] Standard emission factor for cement clinker given in Implementing Regulation Annex II Section B.9.2.2., which states that as a minimum requirement to determine the emission factor, a standard value of 0,525 t CO_2/t cement clinker is applied.

[23] Biomass being the biodegradable fraction in municipal waste. If the emission factor and/or NCV for municipal waste is not known, the standard values in Implementing Regulation Annex II, Section G, Table 2 should be used, which are 11,6 GJ/t and 100 t CO_2/TJ.

In the Table 4-1 above, step 1 is to calculate and attribute direct and indirect emissions associated with the production of cement clinker in the reporting period and derive SEE values for the amount of clinker produced.

Note that the emission factor used for raw meal above is a standard emission factor given in section B.9.2.2 of Annex II to the Methodology Act, which states that, as a minimum requirement to determine the emission factor, a standard value of 0,525 t CO_2/t cement clinker is applied. For using more accurate values, chemical analyses of the raw meal would have to be performed.

Note also that the direct emissions associated with the biomass content of the household waste is calculated separately and is deducted from the total direct emissions. This is because the biodegradable fraction of municipal waste (given as 15% above) is treated as biomass and is effectively zero-rated for total emissions, as the RED II sustainability criteria do not apply to household / municipal waste.

The emission factor used for electricity consumed is the default value for indirect emissions of goods, as provided in Annex II to the Default Values Act. In the example above, the emission factor applied to electricity consumed in the production of cement clinker is the value provided for Lebanon. As an operator, you shall select the emission factor relevant for country where the installation is located.

Table 4-2: Calculation of total direct and indirect SEE values for cement final product (Step 2)

Portland cement production			**Comment**
tonnes clinker / tonne cement ratio	0,95		This is the CCR for Portland cement. The CCR is specific to the cement product produced.
	MWh/t cement	**t CO_2/t**	
Additional electricity consumption	0,085	0,05321	For the cement grinding production process. Calculated as MWh/t x EF for electricity.
Step 2: SEE values are derived for final cement product including the embedded emissions from the precursor cement clinker			
Cement	SEE **Direct**	SEE **Indirect**	
	t CO_2 / t cement	t CO_2 / t cement	
Contribution of precursor (clinker)	0,7852	0,0387	Calculated using the CCR, for example for SEE direct as 0,8265 t CO_2 / t x 0,95 = 0,7852 t CO_2 / t
Production process		0,05321	As above
Total specific embedded emissions	**0,7852**	**0,0919**	Sum of SEEs
Total (direct + indirect) SEE	**0,8771**		Value for which CBAM certificates need to be surrendered, before subtraction of free allocation adjustment (and carbon price paid in third country)

4.2 Calculation using bubble approach

Here, we only outline the different path of the calculation. The explanation for the values chosen can be found in section 4.1.

Table 4-3: Calculation of total direct and indirect SEE values for cement using the bubble approach

Activity level			
Total production of clinker:	1 255 000	t clinker / yr	
Clinker to cement ratio (CCR)	0,950		
Total production of cement:	1 321 053	t cement / yr	
expressed as functional unit:	1 255 000	t clinker / yr	
Emissions / Cement:	**direct**	**indirect**	
total bubble's emissions:	1 037 310	121 359	t CO_2 / yr
SEE per functional unit	0,8265	0,0967	t CO_2 / t clinker
SEE per tonne of cement:	0,7852	0,0919	t CO_2 / t cement

4.3 Calculation of free allocation adjustment and CBAM obligation

As the operator is monitoring and reporting actual data for the cement produced in its installation, also the free allocation adjustment can be composed. The calculation is outlined below:

- CBAM BM for cement = 0 (Column A of the Table in the Annex to the Free Allocation Adjustment Act)[24,25]
  - CBAM BM for grey clinker = 0,666 (column A)
  - M_i (clinker) = 0,95 (in line with the operator's monitoring)
  - $SEFA_{cement} = 0 + 0{,}95 \times 0{,}666 = 0{,}6327$ t CO_2 / t cement
  - Assumption: Year = 2026.:
    - CSCF = 1,00
    - CBAM = 97,5%
  - Specific embedded free allocation (SEFA**)** = 0,6327 × 0,975 × 1,00 = = **0,617 t CO_2 / t cement**

The total embedded emissions to be reported by the authorised declarant (EU importer) for the import of Portland cement into the EU during the definitive period may then be determined, for example, for the import of **100 tonnes of Portland cement**:

[24] The Free Allocation Adjustment Act provides values for products based on grey cement clinker with indicator "(A)" and white clinker with indicator "(B)". The operator here knows that the product is grey clinker and selects the values accordingly.

[25] In most installations, no direct emissions occur when producing cement using cement clinker. Therefore, the benchmark of the cement production process is zero, while the benchmark of the total process is significantly higher, as it reflects the clinker production's benchmark.

- **Definitive period:**
    - Direct embedded emissions = 100 t × 0,7852 t CO_2 / t = 78,52 t CO_2
    - Indirect embedded emissions = 100 t × 0,0919 t CO_2 / t = 9,19 t CO_2

    Total embedded emissions: **87,71 t CO_2**

    Minus Free Allocation adjustment: **61,7 t CO_2**

    (Further deduction could be made if a carbon price was paid in the country of production – outside the scope of this guidance)

    Total CBAM obligation to be covered by the authorised declarant with CBAM certificates: 87,7 – 61,7 = **26 certificates**

Annex A List of definitions

Term	Definition
'Accuracy'	means the closeness of the agreement between the result of a measurement and the true value of the particular quantity or a reference value determined empirically using internationally accepted and traceable calibration materials and standard methods, taking into account both random and systematic factors
'Activity data'	means the amount of fuels or materials consumed or produced by a process relevant for the calculation-based methodology, expressed in terajoules (TJ), mass in tonnes or (for gases) volume in normal cubic metres, as appropriate
'Actual emissions'	means the emissions calculated based on primary data from the production processes of goods and from the production of electricity consumed during those processes as determined in accordance with the methods set out in Annex IV [to the Methodology Act]
'Activity level'	means the quantity of goods to which the same functional unit applies that are produced within the system boundaries of a production process during a reporting period
'Aggregated goods categories'	means aggregated goods categories pursuant to Table 1 of point 2 of Annex I. A group of CBAM goods with their CN codes, used for the purpose of defining system boundaries of production processes.
'Agricultural, aquaculture, fisheries and forestry residues'	means residues that are directly generated by agriculture, aquaculture, fisheries and forestry and that do not include residues from related industries or processing
'Authorised CBAM declarant'	means a person authorised by the competent authority in accordance with Article 17 [of the CBAM Regulation]. The 'authorised CBAM declarant' [which replaces the term 'reporting declarant' from the transitional period] may be importers of CBAM goods, or their indirect customs representatives, who have applied for and been granted the status of authorised CBAM declarant.
'Batch'	means an amount of fuel or material representatively sampled and characterised, and transferred as one shipment or continuously over a specific period of time
'Biomass'	means biomass as defined in Article 2, point (24) of the Renewable Energy Directive (RED II)[26]; it includes bioliquids and biofuels as defined in Article 2, point (32) and (33), biomass fuels as defined in Article 2, point (27) and biogas as defined in Article 2 point (28) of the RED II;

[26] Directive (EU) 2018/2001

Term	Definition
'Biomass fraction'	means the ratio of carbon stemming from biomass to the total carbon content of a fuel or material, expressed as a fraction
'Calculation factors'	means net calorific value, emission factor, preliminary emission factor, oxidation factor, conversion factor, carbon content or biomass fraction
'Calibration'	means the set of operations, which establishes, under specified conditions, the relations between values indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material and the corresponding values of a quantity realised by a reference standard
'Carbon price'	means the monetary amount paid in a third country, under a carbon emissions reduction scheme, in the form of a tax, levy or fee or in the form of emission allowances under a greenhouse gas emissions trading system, calculated on greenhouse gases covered by such a measure, and released during the production of goods
'CBAM certificate'	means a certificate in electronic format corresponding to one tonne of CO_2e of embedded emissions in goods
'CO_2 emission factor'	means the weighted average of the CO_2 intensity of electricity produced from fossil fuels within a geographic area. The CO_2 emission factor is the result of the division of the CO_2 emission data of the electricity sector by the gross electricity generation based on fossil fuels in the relevant geographic area. It is expressed in tonnes of CO_2 per megawatt-hour
'Combined nomenclature' (CN)	means the classification of goods, designed to meet the needs of: i) the Common customs tariff, setting import duties for products imported into the European Union (EU), as well as the Integrated tariff of the European Communities (Taric), incorporating all EU and trade measures applied to goods imported into and exported out of the EU; ii) the international trade statistics of the EU. The CN provides the means of collecting, exchanging and publishing data on EU international trade statistics. It is also used for the collection and publication of international trade statistics in intra-EU trade.[27]
'Combustion emissions'	means greenhouse gas emissions occurring during the exothermic reaction of a fuel with oxygen
'Competent authority'	means the authority designated by each Member State in accordance with Article 11 [of the CBAM Regulation]

[27] For definition see: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Combined_nomenclature_(CN)

Term	Definition
'Continuous emission measurement' (CEM)	means a set of operations having the objective of determining the value of a quantity by means of periodic measurements, applying either measurements in the stack or extractive procedures with a measuring instrument located close to the stack, whilst excluding measurement methodologies based on the collection of individual samples from the stack
'Control system'	means the operator's risk assessment and entire set of control activities, including the continuous management thereof, that an operator has established, documented, implemented and maintained pursuant to point A.2. of Annex II
'Complex goods'	means goods other than simple goods
'Conservative'	means that a set of assumptions is defined in order to ensure that no under-estimation of reported emissions or over-estimation of production of heat, electricity or goods occurs
'Conversion factor'	means the ratio of carbon emitted as CO_2 to the total carbon contained in the source stream before the emitting process takes place, expressed as a fraction, considering CO emitted to the atmosphere as the molar equivalent amount of CO_2
'Co-product'	means any of two or more products resulting from the same production process
'Customs declarant'	means a declarant as defined in Article 5, point (15) of Regulation (EU) No 952/2013 lodging a customs declaration for release for free circulation of goods in its own name or the person in whose name such a declaration is lodged
'CCUS system'	means a group of economic operators with technically connected installations and transport equipment for CO_2 capture, transport, use in production of goods, or geological storage
'Data flow activities'	mean activities related to the acquisition, processing and handling of data that are needed to draft an emissions report from primary source data
'Data set'	means one type of data, either at installation level or production process level as relevant in the circumstances, as any of the following: (a) the amount of fuels or materials consumed or produced by a production process as relevant for the calculation-based methodology, expressed in terajoules, mass in tonnes, or for gases as volume in normal cubic metres, as appropriate, including for waste gases; (b) a calculation factor;

Term	Definition
	(c) net quantity of measurable heat, and the relevant parameters required for determining this quantity, in particular: i) mass flow of heat transfer medium; and ii) enthalpy of transmitted and returned heat transfer medium, as specified by composition, temperature, pressure and saturation; (d) quantities of non-measurable heat, specified by the relevant quantities of fuels used for producing the heat, and the net calorific value (NCV) of the fuel mix; (e) quantities of electricity; (f) quantities of CO_2 transferred between installations; (g) quantities of precursors received from outside the production process, and their relevant parameters, such as country of origin, used production route, specific direct and indirect emissions;
'Default value'	means a value, which is calculated or drawn from secondary data, which represents the embedded emissions in goods
'Direct emissions'	means emissions from the production processes of goods including emissions from the production of heating and cooling that is consumed during the production processes, irrespective of the location of the production of the heating or cooling
'Embedded emissions'	means direct emissions released during the production of goods and indirect emissions from the production of electricity that is consumed during the production processes, calculated in accordance with the methods set out in Annex IV and further specified in the implementing acts adopted pursuant to Article 7(7)
'Emissions'	mean the release of greenhouse gases into the atmosphere from the production of goods
'Emission factor'	means the average emission rate of a greenhouse gas relative to the activity data of a source stream assuming complete oxidation for combustion and complete conversion for all other chemical reactions
'Emission factor' for electricity	means the default value, expressed in CO_2e, representing the emission intensity of electricity consumed in production of goods.
'Emission source'	means a separately identifiable part of an installation or a process within an installation, from which relevant greenhouse gases are emitted
EU ETS	means the system for greenhouse gas emissions allowance trading within the Union in respect of activities listed in Annex I to Directive 2003/87/EC other than aviation activities

Term	Definition
'EU ETS allowance'	means an allowance as defined in Article 3, point (a), of Directive 2003/87/EC in respect of activities listed in Annex I to that Directive other than aviation activities
'Fossil carbon'	means inorganic and organic carbon that is not biomass
'Fossil fraction'	means the ratio of fossil carbon to the total carbon content of a fuel or material, expressed as a fraction
'Fugitive emissions'	means irregular or unintended emissions from sources that are not localised, or too diverse or too small to be monitored individually
'Functional unit'	means the reference unit used for the calculation of embedded emissions in goods
'Goods'	means goods listed in Annex I [to the CBAM the CBAM Regulation]
'Greenhouse gases'	mean greenhouse gases as specified in Annex I [to the CBAM the CBAM Regulation] in relation to each of the goods listed in that Annex
'Heat meter'	means a thermal energy meter or any other device to measure and record the amount of thermal energy produced based upon flow volumes and temperatures
'Importer'	means either the person lodging a customs declaration for release for free circulation of goods or a bill of discharge in accordance with Article 175(5) of Delegated Regulation (EU) 2015/2446 in its own name and on its own behalf or, where the customs declaration is lodged by an indirect customs representative in accordance with Article 18 of Regulation (EU) No 952/2013, the person on whose behalf such a declaration is lodged
'Importation'	means release for free circulation as provided for in Article 201 of Regulation (EU) No 952/2013
'Indirect customs Representative'	A customs representative appointed by an importer established in a Member State. If the importer appoints an indirect customs representative and they agree, they can act as an authorised CBAM declarant and submit the application for an authorisation on behalf of the importer. The indirect customs representative is liable for all operations made for the importers it represents
'Indirect emissions'	means emissions from the production of electricity which is consumed during the production processes of goods, irrespective of the location of the production of the consumed electricity
'Inherent CO_2'	means CO_2 which is part of a source stream.
'Installation'	means a stationary technical unit where a production process is carried out

Term	Definition
'Legal metrological control'	means the control exercised by a public authority or regulator of the measurement tasks intended for the field of application of a measuring instrument, for reasons of public interest, public health, public safety, public order, protection of the environment, the levying of taxes and duties, the protection of consumers and fair trading
'Measurable heat'	means a net heat flow transported through identifiable pipelines or ducts using a heat transfer medium, such as, in particular, steam, hot air, water, oil, liquid metals and salts, for which a heat meter is or could be installed
'Measurement point'	means the emission source for which continuous emission measurement systems (CEMS) are used for emission measurement, or the cross-section of a pipeline system for which the CO_2 flow is determined using continuous measurement systems
'Measurement system'	means a complete set of measuring instruments and other equipment used to determine variables for the monitoring and calculation of emissions
'Minimum requirements'	means monitoring methods using the minimum efforts allowed for determining data in order to result in emission data acceptable for the purpose of the CBAM Regulation.
'Mixed fuel'	means a fuel which contains both biomass and fossil carbon
'Mixed material'	means a material which contains both biomass and fossil carbon
'Multifunctional process'	means a process that delivers multiple outputs or whose outputs feed into several production processes
'Net calorific value' (NCV)	means the specific amount of energy released as heat when a fuel or material undergoes complete combustion with oxygen under standard conditions, less the heat of vaporisation of any water formed
'Non-CBAM good'	means any good produced in the installation that is not included in Annex I to the CBAM Regulation
'Non-measurable heat'	means all heat other than measurable heat
'Operator'	means any person who operates or controls an installation in a third country, including a parent company that controls an installation in a third country
'Oxidation factor'	means the ratio of carbon oxidised to CO_2 as a consequence of combustion to the total carbon contained in the fuel, expressed as a fraction, considering carbon monoxide (CO) emitted to the atmosphere as the molar equivalent amount of carbon dioxide (CO_2)
'Power purchase agreement'	means a contract under which a person agrees to purchase electricity directly from an electricity producer

Term	Definition
'Precursor'	means any input material into a production process included in the list of goods set out in Annex I to the CBAM Regulation;
'Preliminary emission factor'	means the assumed total emission factor of a fuel or material based on the carbon content of its biomass fraction and its fossil fraction before multiplying it by the fossil fraction to produce the emission factor
'Production process'	means the chemical and physical processes carried out to produce goods in an installation
'Production route' [28]	means a specific technology used in a production process to produce goods
'Process emissions'	means greenhouse gas emissions other than combustion emissions occurring as a result of intentional and unintentional reactions between substances or their transformation, for a primary purpose other than the generation of heat, including from the following processes: (a) the chemical, electrolytic or pyrometallurgical reduction of metal compounds in ores, concentrates and secondary materials; (b) the removal of impurities from metals and metal compounds; (c) the decomposition of carbonates, including those used for flue gas cleaning; (d) chemical syntheses of products and intermediate products where the carbon bearing material participates in the reaction; (e) the use of carbon containing additives or raw materials; (f) the chemical or electrolytic reduction of metalloid oxides or non-metal oxides such as silicon oxides and phosphates.
'Proxy data'	means annual values which are empirically substantiated or derived from accepted sources and which an operator uses to substitute a data set[29] for the purpose of ensuring complete reporting
'Rebate'	means any amount that reduces the amount due or paid by a person liable for the payment of a carbon price, before its payment or after, in a monetary form or in any other form.
'Recommended improvements'	means monitoring methods which are proven means to ensure that data are more accurate or less prone to mistakes than by mere application of minimum requirements
'Related emissions'	means: – the direct emissions caused by the source stream or emission source concerned; – emissions attributed to a quantity of measurable heat; – the indirect emissions related to the quantity of electricity concerned; – embedded emissions of a material produced or of a precursor consumed.

[28] Note that different production routes can fall within the same production process.

[29] Refers to the activity data or the calculation factors.

Term	Definition
'Reporting period'	means the period corresponding to the calendar year during which the good was produced and used by the authorised CBAM declarant as reference for the determination of embedded emissions
'Residue'	means a substance that is not the end product that a production process directly seeks to produce; it is not a primary aim of the production process and the process has not been deliberately modified to produce it
'Simple goods'	means goods produced in a production process requiring exclusively input materials (precursors) and fuels having zero embedded emissions
'Source stream'	means either of the following: (a) a specific fuel type, raw material or product giving rise to emissions of relevant greenhouse gases at one or more emission sources as a result of its consumption or production; (b) a specific fuel type, raw material or product containing carbon and included in the calculation of greenhouse gas emissions using a mass-balance method
'Specific embedded emissions'	means the embedded emissions of one tonne of goods, expressed as tonnes of CO_2e emissions per tonne of goods
'Standard conditions'	means temperature of 273,15 K and pressure conditions of 101 325 Pa defining normal cubic metres (Nm^3)
'System boundary'	means the group of chemical or physical processes included in the calculation of embedded emissions of goods under the same aggregated goods category. The system boundaries shall cover direct emissions, indirect emissions for goods not listed in Annex II to the CBAM Regulation, and the embedded emissions of any precursor.
'Third country'	means a country or territory outside the customs territory of the [European] Union
'Tonne of $CO_2(e)$'	means one metric tonne of carbon dioxide ('CO_2'), or an amount of any other greenhouse gas listed in Annex I [to the CBAM Regulation] with an equivalent global warming potential ['CO_2e']
'Transmission system operator'	means an operator as defined in Article 2, point (35) of Directive (EU) 2019/944 of the European Parliament and of the Council ([30]).

[30] Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity and amending Directive 2012/27/EU (OJ L 158, 14.6.2019, p. 125).

Term	Definition
'Uncertainty'	means a parameter, associated with the result of the determination of a quantity, that characterises the dispersion of the values that could reasonably be attributed to the particular quantity, including the effects of systematic as well as of random factors, expressed in percent, and describes a confidence interval around the mean value comprising 95% of inferred values taking into account any asymmetry of the distribution of values
'Waste'	means any substance or object which the holder discards or intends or is required to discard, excluding substances that have been intentionally modified or contaminated in order to meet this definition
'Waste gas'	means a gas containing incompletely oxidised carbon in a gaseous state under standard conditions which is a result of any of the processes listed under 'process emissions'

Annex B List of abbreviations

Abbreviation	Full term
AD	Activity Data
AEM	Anode Effect Minutes
AEO	Anode Effect Overvoltage
AL	Activity Level
AOD	Argon Oxygen Decarburisation.
BAT	Best Available Techniques
BF	Biomass Fraction
BFG	Blast Furnace Gas
BM	Benchmark
BOF	Basic Oxygen Furnace
BOFG	Basic Oxygen Furnace Gas
BREFs	Best Available Techniques Reference documents
CA	Competent Authority
CBAM	Carbon Border Adjustment Mechanism, the
CCR	Clinker to Cement Ratio
CCS	Carbon Capture and Storage
CCU(S)	Carbon Capture and Utilisation (and Storage)
CEMS	Continuous Emission Measurement Systems
CF	Conversion Factor
CFP	Carbon footprint of products
CHP	Combined Heat and Power
CKD	Cement Kiln Dust
CN	Combined nomenclature
COG	Coke Oven Gas
DRI	Direct Reduced Iron
EAF	Electric Arc Furnace
EEA	European Economic Area (EU-27, Iceland, Norway, Liechtenstein)
EF	Emission Factor
EFTA	European Free Trade Area
EORI	Economic Operator Registration and Identification
ETS	Emissions Trading System
EU ETS	EU Emissions Trading System
EUA	EU Allowances (used in the EU ETS)
EUR	Euro (currency)

Abbreviation	Full term
FAR	Free Allocation Rules (Regulation 2019/331)[31]
FIFO	First-In-First-Out principle
GHG	Greenhouse Gas
GWP	Global Warming Potential
HBI	Hot Briquetted Iron
HS	Harmonised System (for international trade)
IEA	International Energy Agency
ISO	International Organization for Standardization
LULUCF	Land-use, land change and forestry (criteria)
MP	Monitoring Plan
MRR	Monitoring and Reporting Regulation (Regulation 2018/2066)[32]
MRV	Monitoring, Reporting and Verification
MS	Member State(s)
MWh	Megawatt-hour
NCV	Net Calorific Value
NPI	Nickel pig iron
OF	Oxidation Factor
PCI	Pulverised Coal Injection
PEMS	Predictive Emission Monitoring System
PFC	Perfluoro-carbon
PoS	Proofs of Sustainability
RED II	Renewable Energy Directive, re-cast
SEE	Specific embedded emissions
SEFA	Specific embedded free allocation
TARIC	Integrated Tariff of the European Union database
TJ	Terajoules
TSO	Transmission System Operator
UCC	Union Custom Code
UN/LOCODE	United Nations Code for Trade and Transport Location

[31] Free Allocation Rules (Commission Delegated Regulation (EU) 2019/331 of 19 December 2018 determining transitional Union-wide rules for harmonised free allocation of emission allowances pursuant to Article 10a of Directive 2003/87/EC of the European Parliament and of the Council)

[32] Monitoring and Reporting Regulation (Commission Implementing Regulation (EU) 2018/2066 of 19 December 2018 on the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC of the European Parliament and of the Council and amending Commission Regulation (EU) No 601/2012